Two-chirality integral rigidity beyond ordinary Seiberg–Witten type

23 September 2026. This note applies the existing quantitative functional pencil to Bauer–Furuta detected connected sums. The new application is an actual family with even positive and negative Betti numbers and no almost-complex structure in either orientation. The differential estimate is inherited; the actual gauge supply, geography, topology and application are proved together below. No claim of exhaustive literature priority is made.

1. The analytic input and the distinct gauge category

Use tensor norms for Ricci and the Hilbert–Schmidt norm on the Weyl endomorphism of two-forms. For a closed connected oriented smooth four-manifold set

$$T_{\pm} = 2\chi \pm 3\tau, \qquad \mathscr{E}(g) = \inf_{\lambda \neq 0} \frac{\int |\operatorname{Ric}_g - \lambda g|^2}{\lambda^2 \operatorname{Vol}_g(M)}.$$

The actual-class version of the existing two-chirality theorem, Section 2, says: if there is a number $A > 0$ such that every metric admits an actual conformally solvable Spin^c structure satisfying

$$(a_g^+)^2 \geqslant A, \qquad a_g^2 \geqslant T_+, \qquad h = 1 - T_+/A \in (0, 1/3),$$

then a universal constant $c > 0$ gives

$$\mathscr{E}(g) + (U - F(h))_+ \geqslant ch^2, \qquad U = \frac{3\min(T_+, T_-)}{A}, \qquad F(h) = \frac{(4 - \sqrt{1-3h})^2}{9}. \tag{1}$$

The constant is independent of the number of ordinary basic classes, the rank of cohomology, and the virtual dimension of the chosen structure. The only nonclassical analytic premise in this note is (1), whose proof uses the functional-pencil theorem and energy-envelope transfer.

The following source theorem supplies actual structures where ordinary Seiberg–Witten invariants vanish. For two or three simply connected symplectic four-manifolds X_i with $b_+(X_i) \equiv 3 \pmod 4$, Bauer's nonvanishing theorem gives nonzero Bauer–Furuta invariants for the connected sum of their canonical structures. Adding a manifold with negative-definite intersection form and determinant square $-b_2$ preserves this nonvanishing. Independent canonical and exceptional signs give actual monopole classes. See Ishida–LeBrun, Theorem 9, Corollary 8, Proposition 10 and Corollary 11 in *Curvature, Connected Sums, and Seiberg–Witten Theory*. For two or three positive summands no fourth-summand congruence is needed.

2. A quantitative theorem for decomposable symplectic building blocks

Theorem 1. Let $r \in \{2, 3\}$. Let $X_1, \dots, X_r$ be simply connected symplectic four-manifolds with $b_+(X_i) \equiv 3 \pmod 4$, and suppose

$$A = \sum_{i=1}^{r} c_1^2(X_i) > 0.$$

For nonnegative integers k, m, put

$$M = \left(\#_{i=1}^{r} X_i\right) \# k\overline{\mathbb{CP}}^2 \# m(S^1 \times S^3), \qquad D = 4(r-1) + k + 4m, \qquad h = D/A. \tag{2}$$

If $0 < h < 1/3$, every smooth metric on M satisfies (1), with the same universal constant as in the existing two-chirality theorem. In particular, if

$$U \leqslant F(h) + \frac{c}{2} h^2, \tag{3}$$

then

$$\mathscr{E}(g) \geqslant \frac{c}{2} h^2, \qquad \|g^{-1} \operatorname{Ric}_g - \sigma I\|_{\infty,\mathrm{op}} \geqslant h\sqrt{c/8} \quad (\sigma = \pm 1). \tag{4}$$

Under condition (3), M has no Einstein metric of any sign.

Proof. Canonical classes on the symplectic summands have mod-two Seiberg–Witten invariant one. Write $c_i = c_1(X_i)$ and $\alpha = \sum_i c_i$, using the orthogonal connected-sum splitting. The source theorem gives, for every exceptional sign choice, an actual monopole class

$$a_\epsilon = \alpha + \sum_{j=1}^{k} \epsilon_j E_j, \qquad \epsilon_j \in \{\pm 1\}, \quad E_i E_j = -\delta_{ij}. \tag{5}$$

Here $\epsilon = (\epsilon_1, \dots, \epsilon_k)$. These are first Chern classes of specified Spin^c structures with solutions for every metric, not merely formal characteristic vectors. The handle factors are included in the negative-definite gluing theorem; its proof allows a nontrivial Picard torus on that summand.

Topology and the almost-complex structure on each summand give

$$T_+(M) = A - 4(r-1) - k - 4m = A - D, \qquad a_\epsilon^2 = A - k,$$

$$a_\epsilon^2 - T_+(M) = 4(r - 1 + m) \geqslant 0. \tag{6}$$

In particular these are special classes, with actual virtual dimension $r - 1 + m$. For any metric, averaging (5) over the exceptional signs gives

$$2^{-k} \sum_{\epsilon} (a_{\epsilon,g}^+)^2 = (\alpha_g^+)^2 + \sum_j (E_{j,g}^+)^2 \geqslant \alpha^2 = A. \tag{7}$$

Choose one maximizing structure. The choice can depend on the metric, but the resulting structure solves the unperturbed equations on all metrics and hence on every conformal metric. The same chosen class therefore supplies every positive auxiliary weight in the analytic theorem. Its square is positive here, since $k/A \leqslant h < 1/3$, so the negative-square cost is actually zero. Equations (6)–(7) verify all inputs to (1), proving (1) and (4).

A nonzero Einstein constant makes $\mathscr{E} = 0$, contradicting (4). Ricci-flatness is not a zero-defect case of this definition; instead the actual class and (7) give the signed scalar estimate $\int s_-^2 \geqslant 32\pi^2 A > 0$, which rules it out separately. This proves the final assertion. □

This extension changes the source of the gauge information. Splitting M into two pieces of positive b_+ makes its ordinary Seiberg–Witten invariants vanish by the standard connected-sum vanishing theorem. Actual Bauer–Furuta classes persist nevertheless. Thus ordinary basic classes on M cannot be silently substituted for (5).

3. Why low-signature building blocks do not realize the new strip

This is a useful selection calculation, not a general obstruction to the connected-sum method. Write

$$\rho = \frac{\sum_i \tau(X_i)}{A}, \quad x = k/A, \qquad h = \frac{4(r-1) + k + 4m}{A}.$$

The exact opposite-orientation budget is

$$\frac{3T_-}{A} = 3(1 - 6\rho) - 3h + 18x. \tag{8}$$

For zero-signature building blocks and $h < 1/3$, the signature of M is $-k$; hence $U = 3T_+/A = 3(1-h) > 2$, while $F(h) < 16/9$. They cannot approach $U = F(h)$ in this range. This explains why simply taking several product-like inputs does not realize the desired two-chirality strip.

For positive signature, (8) instead shows what must change: the building blocks need a sufficiently positive ratio ρ, and handle addition must supply enough of the surgery budget. Both are realized in the following construction.

4. An actual infinite family with no almost-complex structure

Braungardt–Kotschick's geography theorem realizes every sufficiently large lattice point strictly below any fixed slope less than nine by simply connected minimal symplectic manifolds: Theorem 4 of *Einstein metrics and the number of smooth structures on a four-manifold*. Take its parameter $\epsilon = 1/20$. Consequently, for every sufficiently large integer n, choose such a manifold X_n with

$$\chi_h(X_n) = 20n, \qquad c_1^2(X_n) = 178n. \tag{9}$$

Indeed $178n \leqslant (9 - 1/20)20n - c(1/20) = 179n - c(1/20)$ for large n. Here $\chi_h = (\chi + \tau)/4$ is the topological Todd number for a symplectic manifold; no complex structure on X_n is assumed. The two defining topological equations imply

$$\chi(X_n) = 62n, \quad \tau(X_n) = 18n, \quad b_+(X_n) = 40n - 1 \equiv 3 \pmod 4, \quad b_-(X_n) = 22n - 1. \tag{10}$$

Take two copies of X_n. Set

$$A_n = 356n, \quad h_0 = \frac{19}{300}, \quad F_0 = F(h_0) = \frac{961}{900}, \quad u_0 = \frac{105}{89},$$

$$x_0 = \frac{781}{180225}, \qquad y_0 = \frac{42533}{2883600}. \tag{11}$$

These positive rational numbers satisfy

$$x_0 + 4y_0 = h_0, \qquad u_0 + 15x_0 - 12y_0 = F_0. \tag{12}$$

Define the even nonnegative surgery counts, for sufficiently large n,

$$k_n = 2\left\lfloor \frac{x_0 A_n}{2} \right\rfloor, \qquad m_n = 2\left\lfloor \frac{y_0 A_n}{2} \right\rfloor - 4,$$

$$M_n = X_n \# X_n \# k_n \overline{\mathbb{CP}}^2 \# m_n (S^1 \times S^3). \tag{13}$$

Theorem 2 (realized Bauer–Furuta strip). There exists n_0 such that, for every $n \geqslant n_0$ and every smooth metric on M_n,

$$\boxed{\mathscr{E}(g) \geqslant \frac{ch_0^2}{8} > 0, \qquad \delta_\sigma(g) \geqslant h_0\sqrt{c/32} \quad (\sigma = \pm 1).} \tag{14}$$

The manifolds M_n are pairwise nonhomeomorphic, have vanishing ordinary Seiberg–Witten invariants, and admit no almost-complex structure in either orientation. Both strict Hitchin–Thorpe inequalities hold. Their numerical data lie on the allowed side of the classical optimized energy envelope, but inside its new quantitative exclusion strip for large n.

Proof. Connected-sum topology gives

$$\chi(M_n) = 124n - 2 + k_n - 2m_n, \qquad \tau(M_n) = 36n - k_n,$$

$$T_+ = A_n - 4 - k_n - 4m_n, \quad T_- = 140n - 4 + 5k_n - 4m_n. \tag{15}$$

Write $k_n = x_0 A_n - \alpha_n$ and $m_n = y_0 A_n - \beta_n - 4$, with $0 \leqslant \alpha_n, \beta_n < 2$. Since $\tau(M_n)/A_n \geqslant 9/89 - x_0 = 196/2025 > 0$, the smaller chirality is T_-. Thus, exactly,

$$h_n = h_0 - \frac{\alpha_n + 4\beta_n + 12}{A_n}, \qquad U_n = F_0 + \frac{-15\alpha_n + 12\beta_n + 36}{A_n}. \tag{16}$$

For large n, $0 < h_n < h_0$. The numerator of the correction in U_n lies strictly between 6 and 60. Also

$$F'(h) = \frac{4 - \sqrt{1 - 3h}}{3\sqrt{1 - 3h}}, \qquad 0 < F'(h) \leqslant 31/27 < 2 \quad (0 \leqslant h \leqslant h_0).$$

Since $\alpha_n + 4\beta_n + 12 < 22$, it follows that

$$\boxed{0 < U_n - F(h_n) < \frac{104}{A_n}, \qquad h_n \longrightarrow h_0.} \tag{17}$$

Theorem 1 applies by (10). Choose n_0 so that $h_n \geqslant h_0/2$ and $104/A_n \leqslant ch_0^2/8$. Then (1) implies $\mathscr{E} \geqslant ch_0^2/4 - ch_0^2/8$, proving (14).

The Betti numbers are

$$b_1(M_n) = m_n, \qquad b_+(M_n) = 80n - 2, \qquad b_-(M_n) = 44n - 2 + k_n. \tag{18}$$

They are all even. An almost-complex four-manifold necessarily has $(\chi + \tau)/4 \in \mathbb{Z}$, equivalently $1 - b_1 + b_+$ even. Equation (18) makes this number odd. Reversing orientation replaces b_+ by the also-even b_-, so the same contradiction holds there. No almost-complex structure exists in either orientation. The positive and negative summands in (18) also allow the ordinary connected-sum vanishing argument in both orientations. The second Betti number $b_2 = 124n - 4 + k_n$ is strictly increasing, which gives pairwise distinct homeomorphism types even without fixing the orientation.

Finally $T_+/A_n \to 281/300 > 0$ and $T_-/A_n \to 961/2700 > 0$, proving strict Hitchin–Thorpe. By (17), the classical Einstein energy interval $[F(h_n), U_n]$ has positive length. Both Weyl energies can be positive inside it. It is the quantitative separation, not the classical scalar–Weyl numerical inequalities alone, that eliminates this interval. □

5. The precise increase in scope

1. These are not merely new choices of a complex general-type base in the earlier application. A Kähler surface has odd $b_+ = 1 + 2p_g$, and blowups and $S^1 \times S^3$ additions preserve b_+. Both b_+ and b_- in (18) are even, so neither orientation of M_n can be diffeomorphic to any member of that former category.
2. Ordinary Seiberg–Witten invariants on the whole manifold vanish. The application genuinely uses actual classes retained by stable cohomotopy. It is not obtained by declaring arbitrary cohomology vectors to be monopoles. The virtual dimension is $1 + m_n$, not zero.
3. The previously recorded universal one-third estimate has $\varepsilon_0 \leqslant 1/4$, whereas $1/3 - h_n > 1/3 - h_0 = 27/100$. It does not itself force positive Einstein defect on this family. The classical connected-sum nonexistence threshold $D \geqslant A/3$ likewise does not apply.
4. The new differential input is still the existing functional pencil. Bauer's nonvanishing and Braungardt–Kotschick geography are classical inputs. This note supplies a realized extension of the quantitative obstruction, not a new Bauer–Furuta theorem or new symplectic geography.
5. These manifolds cannot be Yau26 counterexamples: (14) forbids the required arbitrarily tight pointwise inputs. The result establishes a further category of no-Einstein smooth types on which the original pointwise hypotheses are uniformly impossible.

For large n, $\pi_1(M_n)$ is the infinite free group of rank m_n. Myers' theorem already excludes strictly positive Ricci curvature on these objects. The positive-sign pointwise corollary is therefore not a second independent discovery; the substantive extension is the uniform all-metrics integral gap and its negative-sign pointwise consequence, supplied by the actual Bauer–Furuta classes.

The threshold n_0 depends on the inherited analytic constant and the geography existence threshold. No explicit numerical value is claimed. We choose one available X_n at each lattice point. Although

the source geography gives infinitely many smooth structures there, this note does not assert that all remain pairwise distinct after the new connected sum.

`check_line1_bauer_furuta_geography.py` checks the exact topology, rational parameters, parity obstruction, sign of the floor errors, and comparison with the old recorded band. Those checks do not prove gauge nonvanishing, the geography theorem or the inherited analytic theorem.